Experiment 4

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Subject Name: Full Stack Development-II **Subject Code: 23CSH-309**

1. Aim:

To optimize the performance of the EcoTrack React application using memoization techniques and code splitting, and to enhance the user interface using enterprise-grade Material UI components.

2. Objective: After completing this experiment, the student will be able to:

- Understand the causes of unnecessary re-renders in React applications
- Optimize React components using React.memo to prevent avoidable re-renders
- Apply useMemo to efficiently compute derived data and avoid redundant calculations
- Use useCallback to memoize event handler functions and improve component performance
- Implement lazy loading of components and routes using React.lazy and Suspense
- Reduce initial bundle size and improve application load performance through code splitting
- Enhance the visual appearance and usability of the EcoTrack application using Material UI components
- Design a clean, consistent, and responsive user interface using Material UI layouts and typography

3. Implementation/Code:

- src/App.jsx - Added React.lazy and Suspense for code splitting
- src/components/Header.jsx - Added React.memo, useCallback, and Material UI components

- src/pages/Logs.jsx - Added useMemo for filtering and Material UI components
- src/pages/dashboard.jsx - Added useMemo, useCallback, and Material UI components
- src/pages/login.jsx - Added useCallback and Material UI components
- src/routes/ProtectedRoute.jsx - Added React.memo and cleaned up formatting

App.jsx:

```
import { useMemo, memo, useCallback } from "react";
import { useDispatch, useSelector } from "react-redux";
import { fetchLogs } from "../store/logsSlice";
import { useEffect } from "react";
import {
  Container,
  Card,
  CardContent,
  Typography,
  List,
  ListItem,
  ListItemText,
  Chip,
  Box,
  CircularProgress,
  Alert,
  Grid,
} from "@mui/material";
import TrendingDownIcon from "@mui/icons-material/TrendingDown";
import TrendingUpIcon from "@mui/icons-material/TrendingUp";

const Logs = memo(() => {
  const dispatch = useDispatch();
  const { data, status, error } = useSelector((state) => state.logs);

  useEffect(() => {
```

```
  if (status === "idle") {
    dispatch(fetchLogs());
  }
}, [status, dispatch]);

// Memoized filtered data - only recomputes when data changes
const { highImpactLogs, lowImpactLogs } = useMemo(() => {
  return {
    highImpactLogs: data.filter((log) => log.carbon > 4),
    lowImpactLogs: data.filter((log) => log.carbon <= 4),
  };
}, [data]);

const renderLogList = useCallback(
  (logs, isHighImpact) => (
    <List sx={{ width: "100%" }}>
      {logs.map((log) => (
        <ListItem key={log.id} sx={{ paddingY: 1.5 }}>
          <ListItemText
            primary={log.activity}
            secondary={`Carbon Footprint: ${log.carbon} kg CO₂`}
            primaryTypographyProps={{
              fontWeight: 600,
              color: isHighImpact ? "#d32f2f" : "#2ecc71",
            }}
            secondaryTypographyProps={{
              color: "textSecondary",
            }}
          />
          <Chip
            icon={isHighImpact ? <TrendingUpIcon /> : <TrendingDownIcon />}
            label={`${log.carbon} kg`}
            color={isHighImpact ? "error" : "success"}
            variant="outlined"
            sx={{ marginLeft: 2 }}
          />
        </ListItem>
```

```
      ))}
    </List>
  ),
  []
);

if (status === "loading") {
  return (
    <Container maxWidth="lg" sx={{ paddingY: 4 }}>
      <Box sx={{ display: "flex", justifyContent: "center", paddingY: 4 }}>
        <CircularProgress />
      </Box>
    </Container>
  );
}

if (status === "failed") {
  return (
    <Container maxWidth="lg" sx={{ paddingY: 4 }}>
      <Alert severity="error">
        Error loading logs: {error || "Unknown error"}
      </Alert>
    </Container>
  );
}

return (
  <Container maxWidth="lg" sx={{ paddingY: 4 }}>
    <Box sx={{ marginBottom: 4 }}>
      <Typography
        variant="h4"
        component="h1"
        sx={{
          marginBottom: 3,
          fontWeight: 700,
          color: "#2ecc71",
        }}
```

```
  >
    Carbon Footprint Activities
  </Typography>
</Box>

<Grid container spacing={3}>
  {/* High Impact Activities */}
  <Grid item xs={12} md={6}>
    <Card sx={{ height: "100%", boxShadow: 2 }}>
      <CardContent>
        <Box sx={{ display: "flex", alignItems: "center", gap: 1 }}>
          <TrendingUpIcon sx={{ color: "#d32f2f", fontSize: 28 }} />
          <Typography
            variant="h6"
            sx={{
              fontWeight: 700,
              color: "#d32f2f",
            }}
          >
            High Impact Activities ({highImpactLogs.length})
          </Typography>
        </Box>
        <Typography
          variant="body2"
          sx={{
            color: "textSecondary",
            marginBottom: 2,
            marginTop: 1,
          }}
        >
          Activities with carbon footprint &gt; 4 kg CO₂
        </Typography>
        {highImpactLogs.length > 0 ? (
          renderLogList(highImpactLogs, true)
        ) : (
          <Typography variant="body2" sx={{ color: "textSecondary" }}>
            No high impact activities found.
```

```
        </Typography>
      )}
    </CardContent>
  </Card>
</Grid>

{/* Low Impact Activities */}
<Grid item xs={12} md={6}>
  <Card sx={{ height: "100%", boxShadow: 2 }}>
    <CardContent>
      <Box sx={{ display: "flex", alignItems: "center", gap: 1 }}>
        <TrendingDownIcon sx={{ color: "#2ecc71", fontSize: 28 }} />
        <Typography
          variant="h6"
          sx={{
            fontWeight: 700,
            color: "#2ecc71",
          }}
        >
          Low Impact Activities ({lowImpactLogs.length})
        </Typography>
      </Box>
      <Typography
        variant="body2"
        sx={{
          color: "textSecondary",
          marginBottom: 2,
          marginTop: 1,
        }}
      >
        Activities with carbon footprint ≤ 4 kg CO₂
      </Typography>
      {lowImpactLogs.length > 0 ? (
        renderLogList(lowImpactLogs, false)
      ) : (
        <Typography variant="body2" sx={{ color: "textSecondary" }}>
          No low impact activities found.
```

```
                </Typography>
              )}
            </CardContent>
          </Card>
        </Grid>
      </Grid>
    </Container>
  );
});

Logs.displayName = "Logs";

export default Logs;
```

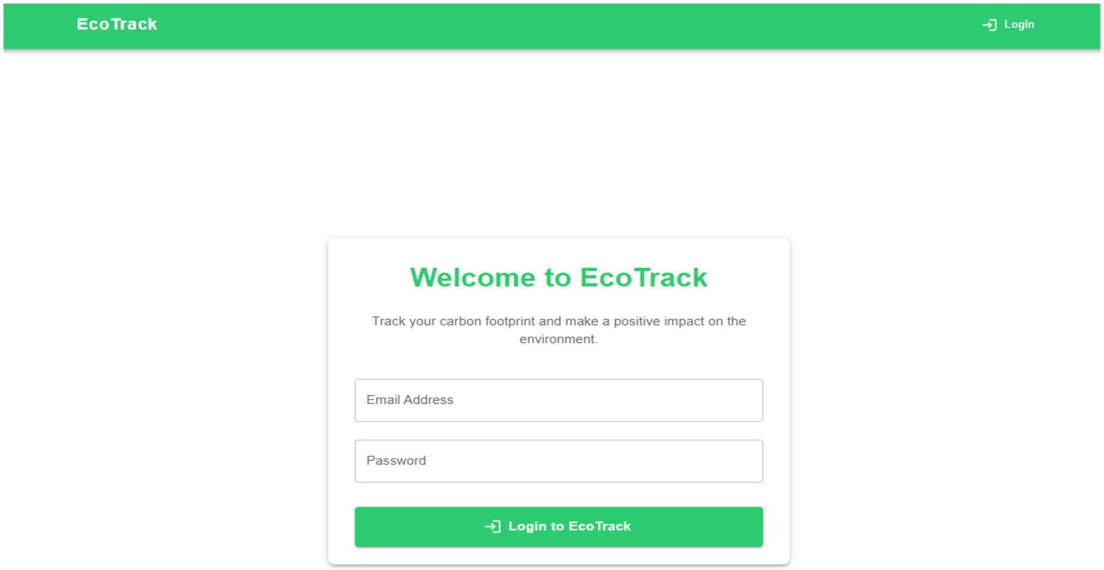
dashboard.jsx:

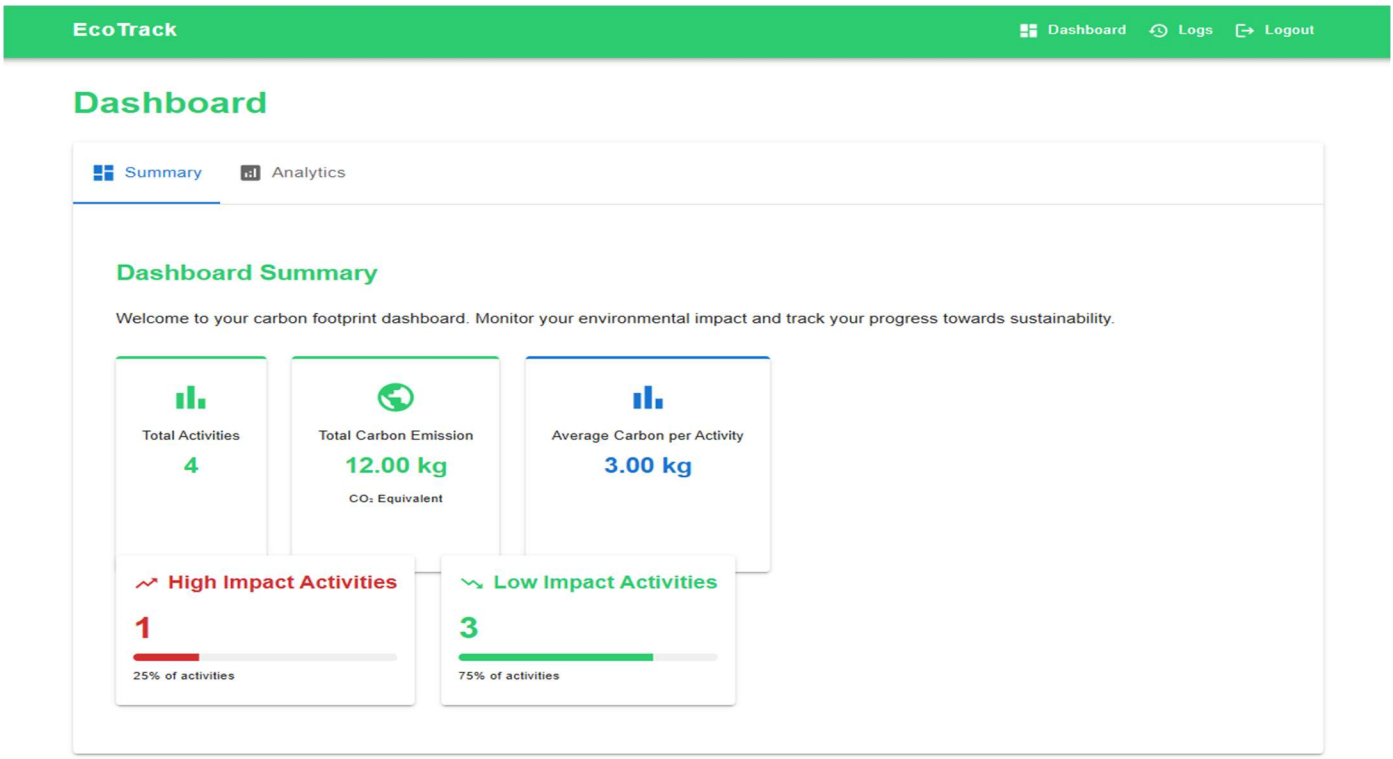
```
import { useDispatch, useSelector } from 'react-redux'; import { useEffect,
useMemo, useCallback } from 'react'; import { fetchLogs } from
'../store/logSlice'; import { Box, Typography, List, ListItem, Button,
CircularProgress, Paper } from
    '@mui/material'; import RefreshIcon from
'@mui/icons-material/Refresh';

const Dashboard = () => {     const dispatch = useDispatch();
const { data: logs, status } = useSelector((state) => state.logs);

   useEffect(() => {        if
(status === 'idle') {
dispatch(fetchLogs());
       }
   }, [status, dispatch]);

   // useCallback memoizes the function to prevent recreation on every render
const handleRefresh = useCallback(() => {        dispatch(fetchLogs());
   }, [dispatch]);
```

```
   if (status === 'loading') {
return (
        <Box sx={{ display: 'flex', justifyContent: 'center', alignItems: 'center', height:
    '50vh' }}>
            <CircularProgress />
        </Box>
      );
   }

   // useMemo prevents recalculating totals on every render
const totalActivities = useMemo(() => {        return
logs.reduce((acc, log) => {            acc[log.activity] =
(acc[log.activity] || 0) + log.carbon;            return acc;
      }, {});
   }, [logs]);

   // useMemo for filtering operations     const highCarbon = useMemo(() =>
logs.filter(log => log.carbon > 4), [logs]);     const lowCarbon = useMemo(() =>
logs.filter(log => log.carbon <= 4), [logs]);

   return (
      <Box sx={{ p: 3 }}>
         <Paper elevation={3} sx={{ p: 3, mb: 3 }}>
            <Typography variant="h4" gutterBottom>Total Activities</Typography>
            <List>
               {Object.entries(totalActivities).map(([activity, carbon]) => (
                  <ListItem key={activity}>
                     <Typography>{activity}: {carbon} Kg CO2</Typography>
                  </ListItem>
               ))}
            </List>
         </Paper>

         <Paper elevation={3} sx={{ p: 2, mb: 2, backgroundColor: '#ffebee' }}>
            <Typography variant="h5" sx={{ color: '#d32f2f', mb: 1 }}>High Carbon
    (&gt; 4 Kg)</Typography>
            <List>
```

```
                {highCarbon.map((log) => (
                    <ListItem key={log.id}>
                        <Typography>{log.activity}</Typography>
                    </ListItem>
                ))}
            </List>
        </Paper>

        <Paper elevation={3} sx={{ p: 2, mb: 3, backgroundColor: '#e8f5e9' }}>
            <Typography variant="h5" sx={{ color: '#2e7d32', mb: 1 }}>Low Carbon (≤ 4
    Kg)</Typography>
            <List>
                {lowCarbon.map((log) => (
                    <ListItem key={log.id}>
                        <Typography>{log.activity}</Typography>
                    </ListItem>
                ))}
            </List>
        </Paper>

        <Button
variant="contained"
color="primary"
            startIcon={<RefreshIcon />}
onClick={handleRefresh}                disabled={status
=== 'loading'}
        >
            {status === 'loading' ? 'Loading...' : 'Refresh Logs'}
        </Button>
    </Box>
  );
};

export default Dashboard;
```

4. Output:


EcoTrack
Login
Welcome to EcoTrack
Track your carbon footprint and make a positive impact on the environment.
Email Address
Password
Login to EcoTrack


EcoTrack
Dashboard
Logs
Logout
Dashboard
Summary
Analytics
Dashboard Summary
Welcome to your carbon footprint dashboard. Monitor your environmental impact and track your progress towards sustainability.
Total Activities
4
Total Carbon Emission
12.00 kg
CO₂ Equivalent
Average Carbon per Activity
3.00 kg
High Impact Activities
1
25% of activities
Low Impact Activities
3
75% of activities

EcoTrack

Dashboard Logs Logout

Dashboard

Summary Analytics

Dashboard Analytics

Detailed analysis of your carbon footprint across different activities.

Activities Ranked by Carbon Impact

Rank	Activity	Carbon Footprint (kg CO_2)	Impact Level	% of Total
#1	Electricity Usage	6	Medium	50.0%
#2	Car Travel	4	Medium	33.3%
#3	Bus Travel	2	Low	16.7%
#4	Cycling	0	Low	0.0%

Highest Impact Activity
6 kg

Lowest Impact Activity
0 kg

Total Activities
4

Total Carbon
12.00 kg

EcoTrack

Dashboard Logs Logout

Carbon Footprint Activities

High Impact Activities (1)

Activities with carbon footprint > 4 kg CO_2

Electricity Usage
Carbon Footprint: 6 kg CO_2 — 6 kg

Low Impact Activities (3)

Activities with carbon footprint ≤ 4 kg CO_2

Car Travel
Carbon Footprint: 4 kg CO_2 — 4 kg

Cycling
Carbon Footprint: 0 kg CO_2 — 0 kg

Bus Travel
Carbon Footprint: 2 kg CO_2 — 2 kg

5 Learning Outcome:

- Identify and reduce unnecessary re-renders in React applications.
- Optimize React components using React.memo, useMemo, and useCallback.
- Implement lazy loading and code splitting to improve application performance.
- Enhance overall React app efficiency through performance optimization techniques.
- Design a clean, responsive user interface using Material UI components.